

Module Mapping — Hi-TGNet → MM-Net

Produced per **Part B** of `methodology_rewrite_prompt_template.md`: re-theme each of the reference paper’s named modules into an equivalent that is faithful to our own architecture and data modality. Nothing here is invented — every module below already exists in `mm_feature_net.py`; what changes is that each is *named* and its role stated, rather than described generically.

Modality note. Hi-TGNet operates on 2-D MRI slices; MM-Net operates on 1-D physiological time series summarised as per-epoch features. The mapping therefore re-themes the *function* of each block, not its geometry — a dilated convolution widening a spatial receptive field maps to a recurrent decoder widening a *temporal* one.

Mapping table

#	Hi-TGNet module		Function in the reference	MM-Net equivalent	Function here
1	Adaptive Block	Spiral	Parallel dilated branches widen the effective receptive field across scales	Nocturnal Context (NCD)	Bidirectional recurrence widens the receptive field from a 30 s epoch to a 20-epoch (10-minute) window, so a stage is read in the context of the night around it
2	Hierarchical Guidance	Tumor	Attention steers representation toward tumour-relevant regions	Direct Cardiorespiratory Pathway (DCP)	Routes the cardiorespiratory stream to the respiratory head without passing through fusion, so oxygen desaturation reaches the decision undiluted by a 13× larger neural vector
3	Uncertainty-aware dual-head predictor		Two heads with confidence scaling on ambiguous cases	Uncertainty-Aware Dual Read-out (UDR)	Staging and respiratory heads from one forward pass, with split-conformal prediction sets giving distribution-free coverage and an explicit abstention
—	<i>(no counterpart)</i>		—	Dual-Physiology Encoder (DPE)	Two parallel encoders keep the neural (188-D) and cardiorespiratory (14-D) representations separate before fusion, which is what makes the modality ablation causally interpretable

Why each name is earned

NCD — the BiLSTM is not decoration. Removing temporal context is the difference between per-epoch classification and hypnogram-shaped output; the HMM decoding layered on top is what makes stage transitions physiologically plausible rather than epoch-independent.

DCP — this is the paper’s actual novel component and the one the ablation isolates. Permutation importance now shows SpO2 costs 0.031 respiratory AUC and -0.001 staging accuracy: the pathway carries exactly what it was built to carry.

UDR — the *only* module for which we have stronger evidence than the reference paper. Hi-TGNet reports an uncertainty-aware head; we report calibrated coverage (0.894 at a 0.90 target) and show that abstention separates hard epochs from easy ones (86.9% vs 59.1% accuracy), with N1 — the stage human scorers agree on least — receiving a singleton set only 19% of the time.

DPE — worth naming because it is load-bearing for the interpretability claim. A single fused encoder would make modality ablation uninterpretable; keeping the streams separate is what licenses the attribution argument.

Propagation required if adopted

Per Part E of the template, adopting these names means updating: the abstract, the bulleted contributions in §I, the methodology subsection headings in §V, the ablation table caption, the architecture figure labels (`mm_architecture_v2.drawio`), and the conclusion. The equations themselves are unchanged — only symbol names would be aligned.

Open decision: the model’s own name

Hi-TGNet takes its acronym from its guiding module (Hierarchical Tumor Guidance $\rightarrow$ Hi-TGNet). "MM-Net" is generic by comparison and does no work for the reader. Options that follow the same pattern:

Candidate	Derivation	Reads as
DCP-Net	Direct Cardiorespiratory Pathway	Names the novel component, exactly parallel to Hi-TGNet
PhysioRoute-Net	Physiologically routed fusion	Names the principle rather than the block
CardioCue-Net	Cardiorespiratory cues driving the second read-out	Clinical, memorable, slightly softer on novelty

This is an authorial decision and I have not made it. DCP-Net is the closest structural parallel to the reference paper, but renaming the model touches every document in the submission, so it should be a deliberate choice rather than a default.